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Real Credit Cycles

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1 Big picture

- **Question.** Can a disciplined form of belief overreaction generate realistic credit cycles—that is, booms with low spreads and high investment followed by busts with rising spreads and falling real activity—without relying on large adverse fundamental shocks?
- **Setting.** The paper embeds *diagnostic expectations* (DE) in a workhorse neoclassical model with heterogeneous firms, risky debt, capital adjustment costs, and aggregate as well as idiosyncratic productivity shocks.
- **Core challenge.** Standard rational-expectations macro-finance models can amplify shocks, but they struggle to produce two joint facts at the same time:
 - survey evidence of systematically overreacting expectations,
 - quantitatively meaningful, countercyclical credit spreads and sharp reversals after good times.
- **Key idea.** Firms and lenders overreact to current TFP news. In good times, both borrowers and lenders become too optimistic. Firms borrow and invest too much, while lenders price debt too aggressively. When news cools off, optimism reverses, credit supply shifts inward, spreads jump, and investment collapses.
- **Main micro evidence.** Managers’ profit forecasts are predictable: high current profits, investment, debt issuance, and forecasts themselves all predict future negative forecast errors. Good times are therefore followed by disappointment.
- **Main micro reversal result.** Predicted disappointment is followed by lower future bond returns and lower future investment growth at the firm level.
- **Main quantitative result.** The estimated diagnosticity parameter is about $\theta \approx 0.913$. The DE model matches realistic belief overreaction, volatile countercyclical spreads, and sharp nonlinear investment responses to modest negative TFP shocks; the RE model does not.
- **Crisis result.** To match the 2007–2009 increase in credit spreads, the DE model requires only a moderate TFP decline close to the measured TFP path. Under the same TFP path, the RE model generates almost no spread increase.
- **Why this paper matters.** The paper gives a micro-data-disciplined behavioral foundation for “financial shocks.” Instead of taking spread shocks or sentiment shocks as exogenous, it shows how they can arise endogenously from overreacting beliefs about fundamentals.

2 Data and measurement (key objects)

2.1 Observed firm-level forecast and balance-sheet panel

- **Manager expectations.** The paper uses the IBES manager guidance database, which records managers’ earnings guidance for their own firms.
- **Core sample window.** Forecast data span roughly 1999–2018.
- **Matched accounting data.** These guidance data are linked to Compustat, which provides profits, investment, liabilities, assets, and other firm financials.

- **Empirical unit.** The core micro regressions are run at the firm–fiscal year level.

2.2 Observed bond-market data

- **Bond issuance data.** The Mergent Fixed Income Securities Database (FISD) contains issue-level information, including spreads at issuance.
- **Secondary-market returns.** TRACE transaction data are used to compute realized corporate bond returns.
- **Bond sample window.** The merged bond sample covers about 2003–2018, because TRACE is needed for realized returns.

2.3 Observed macro data

- **GDP and investment.** U.S. NIPA/BEA data provide aggregate output and private nonresidential fixed investment.
- **Aggregate TFP.** The paper uses Fernald’s TFP series for the crisis-decomposition exercise.
- **Risk-free rate and tax inputs.** External macro/financial sources provide calibration targets such as the risk-free rate and effective corporate tax rates.

2.4 Key measured objects

- **Forecast error.** Realized future profits minus previously forecast profits, scaled by firm assets (or by capital in model-generated data).
- **Investment rate.** Capital expenditures scaled by the firm’s tangible assets or capital stock.
- **Debt.** Long- and short-term liabilities relative to assets in the empirical work; debt is an endogenous state variable in the model.
- **Credit spread.** At the micro level, corporate bond spreads and returns; at the macro level, the paper studies the average spread across firms.
- **Default frequency.** Used as a macro target in the structural estimation.

2.5 Unobservables and timing

- **Idiosyncratic productivity.** Each firm has a persistent micro TFP component z .
- **Aggregate productivity.** All firms are affected by macro TFP A .
- **News states.** Current-period micro and macro shocks, ε_z and ε_A , matter because diagnostic beliefs overweight current news.
- **Timing.** In each period, firms observe current states, decide whether to default, choose labor, investment, equity issuance, and debt issuance, and lenders price debt according to diagnostically distorted default beliefs.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form evidence: predictable forecast errors

The paper first documents nonrationality in manager forecasts using panel regressions of the form

$$ForecastError_{i,t+1} = \alpha_i + \gamma_t + \beta X_{it} + u_{it},$$

where X_{it} is, in turn, current profits, investment, debt, or the manager's own forecast.

- **Interpretation.** Under rational expectations, forecast errors should not be predictable using information known at time t .
- **Finding.** The estimated β 's are negative: high current investment, debt issuance, profits, and forecasts all predict future disappointment.
- **Economic meaning.** Managers are too optimistic in good times and too pessimistic in bad times.

3.2 Reduced-form boom–bust link at the firm level

The paper then studies whether overoptimism predicts future financial and real reversals. The first stage is

$$ForecastError_{i,t+1} = \alpha_i + \gamma_t + \pi Investment_{it} + u_{it},$$

and the second-stage equations are

$$Return_{i,t+1} = a + b \widehat{ForecastError}_{i,t+1} + e_{it},$$

$$\Delta Investment_{i,t+1} = c + d \widehat{ForecastError}_{i,t+1} + controls + v_{it}.$$

- **Interpretation.** High investment today predicts future negative forecast errors, and those predicted disappointments should forecast lower future bond returns and weaker investment growth if beliefs drive boom–bust cycles.
- **Finding.** This is exactly what the data show in Table 2.

3.3 Diagnostic expectations

Let X_{t+1} denote a future variable such as TFP. Under diagnostic expectations, beliefs distort the rational conditional density according to

$$f_t^\theta(X_{t+1} | X_t) \propto f(X_{t+1} | X_t) \left(\frac{f(X_{t+1} | X_t)}{f(X_{t+1} | \mathbb{E}_{t-J}[X_t])} \right)^\theta, \quad (2)$$

with $\theta \geq 0$.

- **Interpretation.** Outcomes that became more likely because of recent news are overweighted in beliefs.

- **Special case.** $\theta = 0$ gives rational expectations.
- **Psychological foundation.** The paper interprets this as selective recall: recent news makes congruent future outcomes more top-of-mind.

3.4 AR(1) productivity and overreacting conditional mean

Suppose aggregate TFP follows

$$\log A_{t+1} = \rho \log A_t + \varepsilon_{t+1}.$$

Then DE imply the distorted expectation

$$\mathbb{E}_t^\theta[\log A_{t+1}] = \rho \log A_t + \theta \rho \sum_{r=0}^{J-1} \rho^r \varepsilon_{t-r}. \quad (3)$$

In the paper's quantitative work, $J = 1$, so current news gets overweighted directly.

- **Interpretation.** Good news leads agents to extrapolate too much; bad news leads them to become too pessimistic.
- **Why DE matter.** Unlike simple exaggerated-persistence stories, optimism can cool off sharply once news recedes into the past, even without a large bad shock.

3.5 Toy spread equation

In the stylized risky-debt setup, if perceived next-period default risk is δ_t^θ , then the promised gross return is

$$\hat{R}_t = \frac{1}{1 - \delta_t^\theta},$$

so the credit spread is

$$S_t = \hat{R}_t - 1 = \frac{\delta_t^\theta}{1 - \delta_t^\theta}. \quad (4)$$

If default occurs when productivity falls below a threshold A^* , then

$$\delta_t^\theta = \Phi\left(\frac{\log A^* - \mathbb{E}_t^\theta[\log A_{t+1}]}{\sigma}\right). \quad (5)$$

- **Interpretation.** Optimistic lenders perceive less default risk and demand lower spreads in good times.
- **Key implication.** Since DE distort perceived future productivity, they directly distort credit supply, not just credit demand.

3.6 Firm technology and capital accumulation in the quantitative model

A representative firm with micro productivity z and macro productivity A produces

$$y = Azk^\alpha n^\nu, \quad \alpha + \nu < 1.$$

Capital evolves as

$$k' = i + (1 - \delta)k,$$

and investment adjustment costs are

$$AC(i, k) = \frac{\eta_k}{2k} i^2.$$

- **Interpretation.** This is a decreasing-returns production environment with costly capital adjustment and one-period time to build.

3.7 Productivity processes

Micro and macro productivity follow AR(1) laws:

$$\log z' = \rho_z \log z + \varepsilon'_z, \quad \varepsilon'_z \sim N(0, \sigma_z^2), \quad (9)$$

$$\log A' = \rho_A \log A + \varepsilon'_A, \quad \varepsilon'_A \sim N(0, \sigma_A^2). \quad (10)$$

Under DE, firms and lenders use distorted transition beliefs:

$$\log z' \mid (\log z, \varepsilon_z) \sim N(\rho_z(\log z + \theta \varepsilon_z), \sigma_z^2), \quad (11)$$

$$\log A' \mid (\log A, \varepsilon_A) \sim N(\rho_A(\log A + \theta \varepsilon_A), \sigma_A^2). \quad (12)$$

- **Interpretation.** The same diagnosticity parameter θ governs overreaction to both idiosyncratic and aggregate news.

3.8 Dividend flow and financing

Firm dividends are

$$d = (1 - \tau)[y - Wn - AC(i, k) - \phi] + q^\theta(s, k', b')b' - i - b + \tau((R - 1)b + \delta k). \quad (13)$$

If $d < 0$, the firm issues equity and pays issuance costs

$$IC(d) = \mathbf{1}(d < 0)(\eta_f + \eta_d|d|).$$

- **Interpretation.** Firms finance investment using retained earnings, risky debt, and if necessary costly external equity.

3.9 Default and continuation values

Let the state be (s, k, b) , where $s = (z, \varepsilon_z, A, \varepsilon_A)$. The firm's value is

$$V^\theta(s, k, b) = \max \left\{ V_D^\theta(s), V_{ND}^\theta(s, k, b) \right\}. \quad (14)$$

If the firm does not default,

$$V_{ND}^\theta(s, k, b) = \max_{k', b', n} \left\{ d - IC(d) + \frac{1}{R} \mathbb{E}^\theta \left[V^\theta(s', k', b') \mid s \right] \right\}. \quad (15)$$

If the firm defaults, it restarts after reorganization with zero debt and assets:

$$V_D^\theta(s) = \frac{1}{R} \mathbb{E}^\theta \left[V^\theta(s', 0, 0) \mid s \right]. \quad (16)$$

- **Interpretation.** Current overoptimism affects not only debt pricing but also the firm's own continuation value and hence its borrowing and investment decisions.

3.10 Debt pricing by risk-neutral lenders

If the lender recovers a fraction γ of the liquidation value after default, debt prices satisfy the zero-profit condition

$$q^\theta(s, k', b') = \frac{1}{R} \mathbb{E}^\theta \left[1 + d_f^\theta(s', k', b') (\mathcal{R}(k', b') - 1) \mid s \right]. \quad (17)$$

The spread relative to the risk-free rate is

$$S^\theta(s, k', b') = \frac{1}{q^\theta(s, k', b')} - R.$$

- **Interpretation.** Lenders' beliefs shift the supply of credit. This is central: DE operate through both the demand for credit and the supply of credit.

3.11 Identification moment for diagnosticity

A linearized version of the model implies that the covariance between forecast-error growth and current changes in investment or debt is

$$\text{Cov} \left[(\pi_{t+2} - \mathbb{E}_{t+1}^\theta \pi_{t+2}) - (\pi_{t+1} - \mathbb{E}_t^\theta \pi_{t+1}), x_t - x_{t-1} \right] = a_\pi a_x \rho \theta (1 + \theta), \quad x \in \{i, b\}. \quad (18)$$

- **Interpretation.** This covariance is positive if and only if $\theta > 0$. It is the paper's key moment condition tying observable forecast-error dynamics to overreaction.

3.12 Estimation

- **Method.** The paper estimates the model by simulated method of moments (SMM).
- **Unknown parameters.** Ten parameters are estimated: $\theta, \rho_z, \sigma_z, \eta_k, \phi, \gamma, \sigma_A, \eta_f, \eta_d, \sigma_\pi$.
- **Targets.** Sixteen moments are targeted: macro moments on average spread, default frequency, and GDP volatility, plus micro moments on variances and covariances of profits, investment, debt, spreads, and forecast-error moments.
- **Model comparison.** The paper also estimates a constrained RE version with $\theta = 0$, allowing a direct disciplined comparison between DE and RE.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in standard RE macro-finance stories

- Standard rational-expectations models with financial frictions can amplify shocks, but they cannot generate the documented predictability of forecast errors.

- Without that belief evidence, they also struggle to match realistic spread volatility and sharp reversals after small negative TFP shocks.
- In the paper’s comparison, the RE model produces almost no aggregate spread volatility relative to the data.

4.2 Why manager forecast data identify overreaction

- Under rational expectations, forecast errors should be orthogonal to information available when the forecast is made.
- The paper instead finds systematic negative predictability of future forecast errors from current profits, investment, debt, and forecasts.
- These moments are especially valuable because they directly discipline the model’s *belief* parameter rather than inferring it only from prices or quantities.

4.3 Why the covariance moments pin down θ

- The key identifying moments are the standard deviation of forecast errors and the covariances of future forecast-error growth with current changes in investment and debt.
- In the model, these covariances are increasing in θ and equal zero under RE.
- This makes overreaction identifiable from micro expectation data, not just from aggregate asset-market moments.

4.4 What identifies the remaining parameters

- ρ_z and σ_z : the persistence and volatility of firm profits and related covariances.
- η_k : investment dynamics and investment volatility.
- ϕ and γ : average default rates and average spreads.
- σ_A : macro GDP volatility.
- η_f and η_d : debt issuance, investment, and spread comovements through the financing margin.
- σ_π : excess profit noise due to accounting or measurement conventions.

4.5 Why the untargeted moments are important

- The paper does not only match targeted moments. It checks whether the estimated DE model can reproduce a set of *untargeted* facts:
 - macro spread volatility,
 - nonlinear fragility after good times,
 - firm- and sector-level boom–bust reversals,
 - the 2007–2009 crisis spread jump.

- These untargeted successes are a central credibility check, because they show that the model is not just overfitting the SMM targets.

4.6 Relation to the literature

- Relative to habit, long-run risk, disaster-risk, or uncertainty-shock models, this paper keeps required returns simple and instead disciplines deviations from rationality using survey-like micro data.
- Relative to rational inattention or exaggerated-persistence stories, DE are attractive because they are microfounded in selective recall and because optimism can reverse sharply even without very large bad shocks.
- Relative to recent DE business-cycle models, this paper adds risky debt, heterogeneous firms, and direct estimation from firm-level expectations.

5 Tables and figures

5.1 Table 1: Predictable forecast errors

Read:

- Current firm conditions predict future forecast errors. This is direct evidence against rational expectations in managerial guidance.
- The coefficients are all negative: profits (-0.044), investment (-0.457), debt (-0.326), and forecasts (-0.244).
- The strongest reduced-form predictor is investment: a 1 standard deviation increase in investment predicts about 10.5 percentage points more disappointment next year.
- The key message is that good times are systematically followed by negative earnings surprises.

5.2 Table 2: Linking forecast errors and firm reversals

Read:

- The first stage shows that higher current investment predicts more negative future forecast errors: coefficient -0.565 , with first-stage $F = 29$.
- In the second stage, predicted disappointment forecasts lower bond returns: coefficient on forecast error is 0.007 .
- Predicted disappointment also forecasts lower future investment growth: coefficient 0.483 .
- So the data show firm-level boom–bust cycles in both financial prices and real activity.

5.3 Table 3 and Table 4: Calibration inputs and estimated parameters

Read:

TABLE 1—PREDICTABLE FORECAST ERRORS

Dependent variable:	Forecast error _{t+1}			
	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Profits _t	−0.044 (0.021)			
Investment _t		−0.457 (0.065)		
Debt _t			−0.326 (0.063)	
Forecast _t				−0.244 (0.031)
Firm effects	✓	✓	✓	✓
Year effects	✓	✓	✓	✓
Years	1999–2018	1999–2018	1999–2018	1999–2018
Firm-years	9,666	9,666	9,666	9,666

Notes: The table reports panel OLS estimates from the merged Compustat-IBES sample of the coefficients of a regression of forecast errors on the indicated variable. The standard errors are clustered at the firm level. All variables are measured at the firm-fiscal year level. Forecast errors in $t + 1$ are realized earnings in $t + 1$ minus firm forecasts in t , scaled by firm tangible assets. Investment in t is capital expenditures, scaled by firm tangible assets. Debt is long-term and short-term liabilities at the end of t , scaled by firm total assets. Forecasts are manager guidance in the year t earnings call about earnings in year $t + 1$, scaled by firm tangible assets. Note that $0.01 = 1\%$ relative to the firm's assets.

TABLE 2—LINKING FORECAST ERRORS AND FIRM REVERSALS

Dependent variable:	Forecast error _{t+1}	Return _{t+1}	Δ Investment _{t+1}
IV stage:	First (1)	Second (2)	Second (3)
Forecast error _{t+1}		0.007 (0.004)	0.483 (0.061)
Investment _t	−0.565 (0.104)		
Year effects	✓	✓	✓
Profit control			✓
Years	2002–2018	2002–2018	2002–2018
Firm-years	2,852	2,852	2,852
First-stage F	29		

Notes: The table reports estimates of specifications on the merged Compustat-IBES-FISD/TRACE sample at the firm-fiscal year level. Column 1 reports IV first-stage estimates, and the remaining columns report IV second-stage estimates. Column 3 controls for current profits in the second stage. Standard errors are clustered at the firm level. Note that $0.01 = 1\%$ relative to the firm's assets.

TABLE 3—EXTERNALLY FIXED PARAMETERS

Parameter	Role	Value	Source
1 δ	Depreciation rate	0.10	Annual solution
2 R	Risk-free rate	1.04	Annual solution
3 α	Capital elasticity	0.25	Bloom et al. (2018)
4 ν	Labor elasticity	0.50	Bloom et al. (2018)
5 ρ_A	Macro TFP autocorrelation	0.95	Bloom et al. (2018)
6 τ	Corporate income tax	0.20	Effective rates, CBO (2017)

Note: The table reports the parameter symbol, numerical value, a description, and source information for each of the externally fixed parameters.

TABLE 4—PARAMETER ESTIMATES

Parameter	Role	DE model	RE model
1 θ	Diagnostics	0.913 (0.097)	—
2 ρ_z	Micro TFP autocorrelation	0.670 (0.024)	0.924 (0.011)
3 σ_z	Micro TFP volatility	0.251 (0.010)	0.096 (0.004)
4 η_k	Capital adjustment cost	4.023 (0.329)	3.490 (0.174)
5 ϕ	Fixed operating cost	0.155 (0.014)	0.075 (0.002)
6 γ	Recovery rate	0.233 (0.102)	0.177 (0.275)
7 σ_A	Macro TFP volatility	0.006 (0.001)	0.008 (0.002)
8 η_f	Equity fixed cost	0.012 (0.004)	0.076 (0.003)
9 η_d	Equity linear cost	0.109 (0.028)	0.100 (0.223)
10 σ_ε	Earnings noise	0.678 (0.172)	0.390 (0.077)

Notes: The table reports point estimates and standard errors for each of the parameters in our SMM estimation of both the DE model and the RE model. The moment covariance matrix is based on firm-level clustering in the micro block and a stationary block bootstrap in the macro block. The moment Jacobian is computed numerically. In the SMM estimation, the weighting matrix is optimal, that is, the inverse of the moment covariance matrix.

TABLE 5—ESTIMATED MODEL FIT

	Moment	Data	DE model	RE model
<i>Panel A. Micro moments</i>				
1	cov(Δ Forecast error $_{t+1}$, Δ Investment $_{t-1}$)	0.003	0.005	0.000
2	cov(Δ Forecast error $_{t+1}$, Δ Debt $_{t-1}$)	0.004	0.010	0.000
3	std. dev (Forecast error $_{t+1}$)	0.305	0.207	0.200
4	std. dev (Profit $_t$)	0.262	0.181	0.197
5	corr (Profit $_t$, Investment $_t$)	0.257	0.423	0.125
6	corr (Profit $_t$, Debt $_t$)	0.120	0.479	0.111
7	corr (Profit $_t$, Spread $_t$)	-0.159	-0.039	0.025
8	std. dev (Investment $_t$)	0.067	0.066	0.040
9	corr (Investment $_t$, Debt $_t$)	0.104	0.807	0.515
10	corr (Investment $_t$, Spread $_t$)	-0.057	-0.071	0.135
11	std. dev (Debt $_t$)	0.112	0.109	0.101
12	corr (Debt $_t$, Spread $_t$)	-0.036	-0.025	0.204
13	std. dev (Spread $_t$)	0.011	0.007	0.006
<i>Panel B. Macro moments</i>				
14	$\mathbb{E}$ (Spread $_t$)	0.029	0.023	0.004
15	$\mathbb{E}$ (Default $_t$)	0.003	0.005	0.002
16	std. dev(Δ GDP $_t$)	0.015	0.013	0.017

Notes: The moments were computed on a sample combining information from the Compustat, IBES Manager Guidance, and FISD/Trace Bond databases. For the micro moments, the forecast error, profit, and investment, series are expressed relative to firm tangible capital stocks, while the debt series is scaled by total assets and the spread is in proportional units. For the macro moments, the mean spread is the average across years of the mean spread across firms in the FISD/Trace Bond-Compustat merged database; mean default is the average across years of the mean default rate across firms in the FISD/Trace-Compustat merged database. The GDP series is annual GDP in chained 2012 dollars.

- Table 3 fixes conventional parameters such as $\delta = 0.10$, $R = 1.04$, capital elasticity $\alpha = 0.25$, labor elasticity $\nu = 0.50$, macro TFP persistence $\rho_A = 0.95$, and tax rate $\tau = 0.20$.
- Table 4 is the core structural result: the estimated diagnosticity parameter is $\theta = 0.913$ with standard error 0.097.
- Relative to RE, the DE model estimates lower micro persistence ($\rho_z = 0.670$ versus 0.924) and higher micro volatility ($\sigma_z = 0.251$ versus 0.096), because DE itself generates apparent persistence and volatility in firm decisions.
- The estimated θ is in the same ballpark as prior evidence from analyst and macro forecast data, which is one of the paper's strongest external-validity checks.

5.4 Table 5: Estimated model fit

Read:

- The DE model fits the forecast-error moments that the RE model necessarily misses: for example, the data have positive covariances between forecast-error growth and lagged changes in investment and debt, while the RE model gives zeros by construction.
- The DE model also gets the sign of key spread correlations right, especially the negative correlation of spreads with profits and investment.
- The RE model produces counterfactually procyclical spreads in the micro moments and cannot deliver realistic belief volatility.
- One miss is that the DE model still produces too much positive comovement of debt with profits and investment, which the authors interpret as a missing debt-friction channel rather than a failure of the belief mechanism itself.

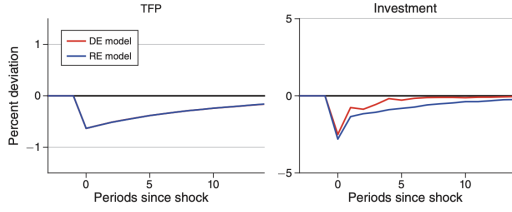


FIGURE 1. INVESTMENT RESPONSE TO A NEGATIVE TFP SHOCK

Notes: The figure plots the simulated generalized impulse responses of investment (right panel) to a 1 standard deviation negative shock to macro TFP (left panel) occurring at period 0. The DE model (red) and RE model (blue) paths are based on identical shocks to TFP over 10,000 simulated experiments following the Koop, Pesaran, and Potter (1996) methodology.

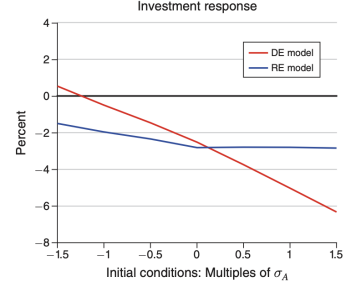


FIGURE 2. INVESTMENT NONLINEARITY

Notes: The vertical axis in the figure reports the simulated first-period impulse response of macro investment to a 1 standard deviation negative shock to macro TFP, and the horizontal axis reports the initial conditions, that is, the magnitude of the shock to macro TFP in the previous period. Both the DE model (red line) and RE model (blue line) are reported on the figure.

5.5 Figure 1 and Figure 2: Normal times versus fragility in good times

Read:

- Figure 1 shows that after a typical negative TFP shock, the DE and RE models have only modestly different average investment responses in normal times.
- Figure 2 is the key state-dependence result: in the DE model, the same negative TFP shock causes much larger investment collapses when it arrives after a positive shock.
- This is the paper's notion of *fragility in good times*. Booms create vulnerable balance sheets and overoptimistic credit pricing, so even modest bad news triggers a sharp reversal.
- The RE model is almost linear in initial conditions and therefore cannot generate this kind of endogenous fragility.

5.6 Figure 3 and Table 6: Financial disruptions and macro spread volatility

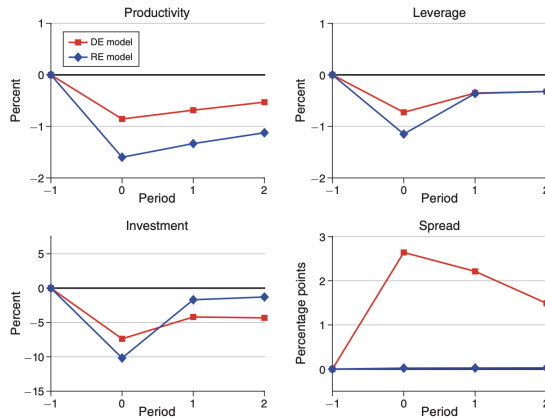


FIGURE 3. FINANCIAL DISRUPTIONS ON AVERAGE

Notes: The figure plots the average path of macro TFP A (top left), macro leverage B/K (top right), macro investment I (bottom left), and the macro spread $\mathbb{E} \text{ Spread}$ (bottom right) around periods of financial disruption from an unconditional simulation of the DE model (red line) versus the RE model (blue line). The variables are plotted in percent (TFP, credit, and investment) or percentage point (spread) deviations from the pre-disruption period. A financial disruption, dated at period t on the horizontal axis, is a period with spread growth in the top 10 percent and investment growth in the bottom 10 percent of realizations in the simulated data for each model.

TABLE 6—MACRO CREDIT SPREAD FLUCTUATIONS

	Standard deviation (1)	Corr. with output (2)
Data	0.011	−0.237
DE model	0.014	−0.563
RE model	0.001	−0.627

Notes: The table reports the standard deviation (column 1) and the correlation with aggregate output (column 2) of the aggregate credit spread. Empirically, output is real GDP from the US NIPA accounts, and the credit spread is the mean across firms in a given year in the merged Compustat-IBES-FISD/TRACE sample. Empirically, output is HP-filtered in logs with smoothing parameter 100. The model moments are computed from an unconditional simulation for the DE model (middle row) and the RE model (bottom row), with an identical set of aggregate TFP shock draws for both model versions. Model quantities refer to the total value (output) or average value (spread) computed from the distribution of firms. Units are in proportional terms; that is, $0.01 = 1\%$.

Read:

- Figure 3 shows that in the DE model, large spread spikes and investment collapses can arise after only mild TFP declines, whereas the RE model needs much larger fundamentals to generate comparable real effects and still fails on spreads.
- Table 6 quantifies the difference: the data’s annual spread volatility is 0.011, the DE model gives 0.014, and the RE model only 0.001.
- Both models generate countercyclical spreads, but only the DE model generates enough *volatility* for the countercyclical to be economically meaningful.
- This is the paper’s main macro-finance success: overreacting beliefs create a realistic investment wedge and volatile credit spreads.

5.7 Table 7 and Table 8: The model reproduces boom–bust reversals

TABLE 7—LINKING FORECAST ERRORS AND MICRO REVERSALS IN THE MODEL

Dependent variable:	Fcst. error _{t+1}	Return _{t+1}	Δ Investment _{t+1}
IV stage:	First (1)	Second (2)	Second (3)
Forecast error _{t+1}		0.011 (0.001)	1.108 (0.004)
Investment _t	−0.860 (0.007)		
Year effects	✓	✓	✓
Profit control			✓
Firms	1,000	1,000	1,000
Firm-years	250,000	250,000	250,000
First-stage <i>F</i>	15,146		

Notes: The table reports first- and second-stage estimates based on simulated firm-level data from the DE model. Column 1 reports the first stage, and columns 2–3 report second-stage regressions. Standard errors are clustered at the firm level. Forecast error is realized minus expected profits normalized by the firm’s capital stock. Investment is the investment rate, that is, capital expenditures normalized by the firm’s capital stock. The bond return is the realized bond return. Column 3 controls for current profits. For all series, 0.01 = 1%.

TABLE 8—LINKING FORECAST ERRORS AND AGGREGATE REVERSALS

Dependent variable:	Forecast error _{t+1}	Return _{t+1}	Δ Investment _{t+1}
IV stage:	First (1)	Second (2)	Second (3)
<i>Panel A. Industry data</i>			
Forecast error _{t+1}		0.047 (0.014)	1.221 (0.050)
Investment _t	−0.476 (0.153)		
Industry effects	✓	✓	✓
Controls		✓	✓
Industries	111	111	111
Industry-years	1,291	1,291	1,291
First-stage <i>F</i>	10		
<i>Panel B. DE model</i>			
Forecast error _{t+1}		0.053 (0.002)	2.039 (0.012)
Investment _t	−0.513 (0.004)		
Industry effects	✓	✓	✓
Controls		✓	✓
Years	50,000	50,000	50,000
First-stage <i>F</i>	17,279		

Notes: The table reports first- and second-stage estimates based on industry data from the Compustat-IBES-FISD/TRACE sample at the SIC3 × fiscal year level (panel A) and simulated aggregate data from the estimated DE model (panel B). Forecast error is average realized minus expected profits normalized by capital stocks. Investment is the average investment rate, that is, capital expenditures normalized by the capital stock. Return is the average realized bond return, and spread is the average realized bond spread relative to the risk-free rate. Column 1 reports the first stage, and columns 2–3 report second-stage regressions. Standard errors are clustered at the industry level. For the purposes of rough comparison with industry data in panel A, simulated aggregate data in panel B is grouped into sequential blocks or industries with the same average length as in the data. So in panel B “Industry effects” refers to fixed effects for these groupings. Columns 2–3 control for the lagged spread and current profits relative to the capital stock. For all series, 0.01 = 1%.

Read:

- Table 7 shows that simulated firm-level data from the DE model reproduce the same qualitative boom–bust logic as in the actual data: high investment predicts future disappointment, and disappointment predicts lower future bond returns and lower investment growth.
- The first-stage coefficient in the model is −0.860, and the second-stage coefficients are 0.011 for returns and 1.108 for investment growth.
- Table 8 shows that the same logic survives aggregation. In industry data, investment predicts future disappointment (−0.476), and forecast errors predict future returns (0.047) and investment growth (1.221). In the DE simulation, the analogous coefficients are qualitatively similar.
- This matters because boom–bust cycles do not wash out in the aggregate. The model delivers sector-level and macro-level reversals too.

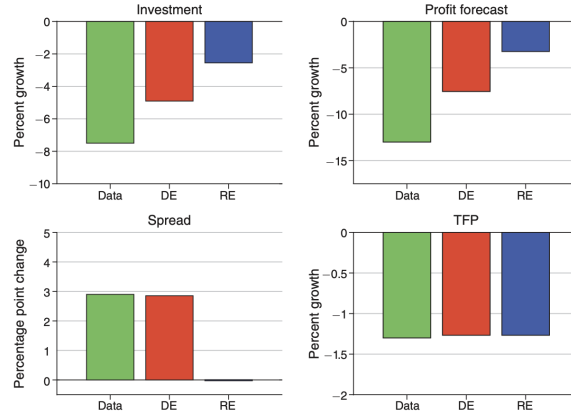


FIGURE 4. THE FINANCIAL CRISIS OF 2007–2009

Notes: Each panel plots a macro series from our crisis decomposition exercise, with the data (green), DE model (red), and RE model (blue) included. The panels plot changes from the precrisis to crisis periods. In the data, all empirical values are averages drawn from the precrisis (2004–2007) or crisis (2008–2009) periods. Spread is the average spread across firms in our Mergent FISD-TRACE-Compustat sample, private nonresidential fixed investment is from NIPA, the macro TFP level is from John Fernald’s website, and profit forecasts are the sum of predicted earnings across all firms in our Compustat-IBES guidance data. In the DE model, we choose the TFP growth series in the bottom right panel in order to exactly match the empirical spread values in the bottom left panel. We feed the resulting TFP growth series into the RE model to produce the RE bar in each panel.

5.8 Figure 4: The financial crisis of 2007–2009

Read:

- The DE model is forced to match the observed jump in spreads from the precrisis to crisis period. Remarkably, the implied TFP decline is close to measured TFP growth, even though TFP was not itself targeted in the fit.
- Under the same TFP path, the RE model produces almost no spread increase.
- The DE model also generates a much larger decline in investment and a larger drop in profit forecasts than the RE model.
- The message is strong: a moderate fundamental shock can generate a major crisis once fragility has been built by overoptimistic beliefs during the boom.

6 Short summary

- **Conclusion.** The paper shows that diagnostic expectations, disciplined by firm-level expectation data, can generate realistic credit cycles and financial reversals.
- **Identification insight.** The key innovation is to use manager forecast-error moments to estimate the strength of belief overreaction directly.
- **Quantitative lesson.** The DE model fits spread volatility, nonlinear investment fragility, firm- and industry-level boom–bust reversals, and the 2007–2009 crisis far better than a comparable RE model.
- **Economic lesson.** “Financial shocks” can arise endogenously when both borrowers and lenders overreact to good news and later reverse.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that a simple, disciplined deviation from rational expectations can materially improve the macro-finance fit of a standard heterogeneous-firm risky-debt model.
- It shows that belief overreaction is not just visible in survey data; it has large equilibrium implications for credit supply, credit spreads, leverage, and investment.
- It also shows that RE plus financial frictions is not enough to explain the joint behavior of expectations, spreads, and real activity in credit cycles.

7.2 Economic intuition

- In good times, firms overestimate future profits and want to borrow more. Lenders also underestimate default risk and are willing to lend too cheaply.
- The boom therefore contains its own fragility: balance sheets become stretched and spreads become too compressed.
- Once news stops improving, both sides reverse. Credit demand falls, credit supply shifts inward, spreads rise, and investment drops sharply.
- This is why modest TFP shocks can generate disproportionately large financial disruptions after booms.

7.3 Takeaway

This paper is best read as a micro-founded behavioral theory of credit cycles. Its contribution is not merely to add sentiment to a business-cycle model, but to discipline the sentiment parameter using firm-level forecast data and then show that this same parameter explains untargeted macro-finance facts. The broader implication is that expectation data should play a much larger role in quantitative macroeconomics, especially when studying episodes in which financial conditions reverse much more sharply than fundamentals alone seem to justify.